

switch on the TV set searching for market information. What strikes me in these morning explorations is the abundance of claims by the alternative medicine vendors of the curing power of their products. These no doubt are caused by the lower advertising rates at that time. To prove their claim, they present the convincing testimonial of someone who was cured thanks to their methods. For instance, I once saw a former throat cancer patient explaining how he was saved by a combination of vitamins for sale for the exceptionally low price of \$14.95 – in all likelihood he was sincere (although of course compensated for his account, perhaps with a lifetime supply of such medicine). In spite of our advances, people still believe in the existence of links between disease and cure based on such information, and there is no scientific evidence that can convince them more potently than a sincere and emotional testimonial. Such testimonial does not always come from the regular guy; statements by Nobel Prize winners (in the wrong discipline) could easily suffice. Linus Pauling, a Nobel Prize winner in chemistry, was said to believe in vitamin C's medicinal properties, himself ingesting massive daily doses. With his bully pulpit, he contributed to the common belief in vitamin C's curative properties. Many medical studies, unable to replicate Pauling's claims, fell on deaf ears as it was difficult to undo the testimonial by a "Nobel Prize winner", even if he was not qualified to discuss matters related to medicine.

Many of these claims have been harmless outside of the financial profits for these charlatans – but many cancer patients may have replaced the more scientifically investigated therapies, in favor of these methods and died as a result of their neglecting more orthodox cures (again, the *non*scientific methods are gathered under what is called "alternative medicine", that is, unproven therapies and the medical community has difficulties convincing the press that there is only one medicine and that alternative medicine is not medicine). The reader might wonder about my claims that the user of these products could be sincere, without it meaning that he was cured by the illusory treatment. The reason is something called "spontaneous remission", in which a very small minority of cancer patients, for reasons that remain entirely speculative, wipe out cancer cells and recover "miraculously". Some switch causes the patient's immune system to eradicate all cancer cells from the body. These people would have been equally cured by drinking

a glass of Vermont spring water or chewing on dried beef as they were by taking these beautifully wrapped pills. Finally, these spontaneous remissions might not be so spontaneous; they might, at the bottom, have a cause that we are not yet sophisticated enough to detect.

The late astronomer Carl Sagan, a devoted promoter of scientific thinking and an obsessive enemy of non-science, examined the cures from cancer that result from a visit to Lourdes in France where people were healed by simple contact with the holy waters, and found out the interesting fact that, of the total cancer patients who visited the place, the cure rate was, if anything, lower than the statistical one for spontaneous remissions. It was lower than the average for those who did not go to Lourdes! Should a statistician infer here that cancer patients' odds of surviving deteriorates after a visit to Lourdes?

PROFESSOR PEARSON GOES TO MONTE CARLO (LITERALLY):
RANDOMNESS DOES NOT LOOK RANDOM!

At the beginning of the twentieth century, as we were starting to develop techniques to deal with the notion of random outcomes, several methods were designed to detect anomalies. Professor Karl Pearson (of Neyman-Pearson fame, familiar to every person who sat in a statistics 101 class) devised the first test of non-randomness (it was in reality a test of deviation from normality, which, for all intents and purposes, was the same thing). He examined millions of runs of what was called a Monte Carlo (the old name for a roulette wheel) during the month of July 1902. He discovered that, with a high degree of statistical significance (with an error of less than one to a billion), the runs were not purely random. What! The roulette wheel was not random! Professor Pearson was greatly surprised at the discovery. But this result in itself tells us nothing; we know that there is no such thing as a pure random draw, for the outcome of the draw depends on the quality of the equipment. With enough minutiae one would be able to uncover the non-randomness somewhere (i.e. the wheel itself may not have been perfectly balanced or perhaps the spinning ball was not completely spherical). Philosophers of statistics call this the *reference case problem*, to explain that there is no true attainable randomness in practice, only in